

Thesis Proposal – Reinforcement Learning Environment for Collectible Card Game Battling



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1 Introduction

Games have long served AI research as a treasure trove of simulations providing varied challenges with relatively well-established rules, tractable scopes of required skills and abundance of human expertise. After deep learning Monte-Carlo-Tree-Search (MCTS) approaches achieved unambiguous superhuman play in go (and similar symmetric perfect information games; shogi and chess), research into structurally complex games has intensified[1]. Over the last few years collectible card games (CCG) have been the focus of various works due to several such structural complexities:

1. enormous state space [2]
2. complex rules defined in natural language [3]
3. non trivial state transitions [4]
4. structurally different but connected game stages [5]
 - a) Deckbuilding
 - b) Battling

For several CCGs researchers have shown that reinforcement learning (RL) agents can outperform expert-crafted classical approaches during Deckbuilding, but these results are based on the usage of simple baselines for Battling. Research into Battling is much less developed and frequently relies on classical agents which are either extremely simple or heavily reliant on expert knowledge. Recently, a team has published a work on joint Deckbuilding and Battling, but to the best of our knowledge that paper[5] is the exemption not the rule; presenting a push into new territory rather than a comprehensive solution. From personal experience in several CCGs we believe that, in any sufficiently complex CCG, Battling and Deckbuilding are linked strongly enough that advanced research on either has to ultimately involve the other. Battling thus not only provides a set of unsolved learning challenges but also enables more informed analysis of previous work on Deckbuilding and consequently is essential for the development of a universal CCG-playing agent.

Contemporary CCG engines are usually high-fidelity, resource-intense implementations intended for human use (or sometimes classical AI) without an explicit API for training of RL agents. A notable exception exists in "Legends of Code and Magic" (LoCM)[6] for which an OpenAI Gym API, capable of running multiple games a second, is implemented. Consequently the majority of contemporary CCG AI research has been conducted in that environment. LoCM, however, aims low in structural complexity, when compared to other CCGs, in favor of an easier learning curve and higher execution speed. It also deviates from environments where human expertise is most abundant, namely preexisting CCGs, making it more difficult to find human experts for qualitative analysis of trained agents.

We believe that other CCGs beyond LoCM provide meaningfully different challenges to self-learning agents, due to their higher structural complexity[7]. Furthermore we hold that the availability of human expertise is crucial for research of explainability in CCG playing; An research question of general interest in RL and of particular value in commercial applications of a CCG agent. We thus propose to implement an CCG environment which offers an efficiently computable, simplified (but congruent) version of a preexisting CCG with a more challenging ruleset. The focus of the proposed implementation shall be on computationally inexpensive sources of structural complexity as well as expandability to enable a gradual approach to the reference CCG. Due to its combat, resource system and redraw mechanic, its wide spread among players and its lacking representation in RL research, we choose Magic: The Gathering (MTG) as our reference CCG.

2 State of Research

CCG research encompasses static analysis (game complexity or card balance) as well as classical and self-learning agents for Deckbuilding and Battling. Deep learning approaches of the last few years have primarily focused on Deckbuilding, relying on relatively simple baselines for Battling. In general such agents are trained for LoCM and do compete against each other instead of humans, but there exists a classical Monte Carlo Tree Search agent[8] which has performed against well human experts in a simplified MTG environment. Another similar exception is the work of Świechowski et al.[9] who report human level Battling for Hearthstone, achieved by their RL agent. Finally, Wei Xi et al.[5] stand out as they consider the joint problem of Deckbuilding and Battling, which is unusual at the time of writing. Overall we identify the following currently under-addressed issues in CCG agent research:

1. Low Interpretability
 - a) Most teams track only winrates
 - b) No focus on explainability of trained agents
 - c) Missing agent-human interaction makes it difficult to notice degenerate/exploitable play patterns[10]
2. LoCM centricity
 - a) LoCM falls on the low end of CCG complexity[7]
 - b) Engines for more complex CCGs are computationally expensive and do not provide standard APIs

3 Proposed Contributions

To address the aforementioned shortcomings we propose to implement a MTG-inspired CCG environment with the following properties:

1. Provides new and complex challenges compared to the LoCM [7]
 - a) Complex Combat: Declared Combat
 - b) Complex Resource System: Resource Cards
 - c) Redraw: MTG-style Mulligan
2. Is of similar computational complexity as LoCM: tens of games per second on a personal pc
3. Provides a standard RL API: Implements the OpenAiGym API
4. Enables Human-Agent "Crossplay"
5. Enables evaluation of expert-defined metrics beyond winrate

References

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